

6. _____ occurs when chromosomes fail to separate properly during meiosis.

Ans: nondisjunction

Difficulty: 2

7. The smaller diploid sister cell that results during meiosis I in females is called _____.

Ans: polar body

Difficulty: 2

8. A _____ is an animal cell with two or more nuclei.

Ans: syncytium

Difficulty: 2

9. A multinucleate plant tissue is called a _____.

Ans: coenocyte

Difficulty: 2

10. The process of matching chromosomes becoming zipped together during the zygotene stage of prophase I is called _____.

Ans: synapsis

Difficulty: 1

Multiple Choice

11. In *Drosophila virilis*, nuclei of somatic cells contain 12 chromosomes while nuclei of sperm cells contain only 6 chromosomes. What does “n” equal for this species?

A) 3

B) 6

C) 12

D) 24

E) 6 or 12, depending on cell type

Ans: B

Difficulty: 1

12. The stage of mitosis when chromosomes condense to form rod-shaped structures visible under the microscope is called:

- A) interphase.
- B) prophase.
- C) metaphase.
- D) anaphase.
- E) telophase.

Ans: B

Difficulty: 1

13. The stage of mitosis when sister chromatids separate from each other and migrate to opposite poles of a cell is called:

- A) interphase.
- B) prophase.
- C) metaphase.
- D) anaphase.
- E) telophase.

Ans: D

Difficulty: 1

14. During which of the following stages of the cell cycle would a chromosome consist of only a single chromatid?

- A) G1
- B) G2
- C) mitotic prophase
- D) mitotic metaphase
- E) all of the above

Ans: A

Difficulty: 1

15. Which of the following statements applies to homologous chromatids?

- A) They are attached to the same centromere
- B) They are genetically identical
- C) They segregate from each other at anaphase of mitosis
- D) They contain the same genes in the same order but may have different alleles of some genes
- E) More than one of the above statements applies to homologous chromatids

Ans: D

Difficulty: 2

16. *Drosophila melanogaster* has four pairs of chromosomes. Sperm from this species are formed by a meiotic process in which homologous chromosomes pair and segregate but do not undergo crossing over. How many genetically different kinds of sperm could be produced by a *Drosophila melanogaster* male?

- A) 4
- B) 8
- C) 16
- D) 64
- E) 256

Ans: C

Difficulty: 3

17. Nondisjunction can occur at either the first or second division of meiosis. XYY individuals would most likely arise from nondisjunction at the _____ meiotic division in the _____.

- A) first, mother
- B) second, mother
- C) first, father
- D) second, father
- E) more than one of the above could give rise to XYY individuals

Ans: D

Difficulty: 3

18. Nondisjunction can occur at either the first or second division of meiosis. XXY individuals could arise from nondisjunction at the _____ meiotic division in the _____.

- A) first, mother
- B) second, mother
- C) first, father
- D) second, father
- E) more than one of the above could give rise to XXY individuals

Ans: E

Difficulty: 3

19. Crossing over between homologous chromosomes occurs at which of the following stages of meiosis?

- A) S phase
- B) prophase I
- C) metaphase I
- D) anaphase I
- E) prophase II

Ans: B

Difficulty: 1

20. Humans have 22 pairs of autosomes plus one sex chromosome pair. How many autosomes are present in a normal human primary spermatocyte?

- A) 22
- B) 23
- C) 44
- D) 45
- E) 46

Ans: C

Difficulty: 2

21. Premeiotic germ cells that divide mitotically in females are referred to as:

- A) primary oocytes.
- B) secondary oocytes.
- C) ootids.
- D) oogonia.
- E) ova.

Ans: D

Difficulty: 1

22. The cells in which meiosis occurs in males are referred to as:

- A) spermatocytes.
- B) spermatogonia.
- C) spermatids.
- D) oocytes.
- E) sperm.

Ans: A

Difficulty: 1

23. A chromosome with two arms of equal length is referred to as:

- A) acrocentric.
- B) homologous.
- C) telocentric.
- D) metazoan.
- E) metacentric.

Ans: E

Difficulty: 1

24. At which of the following stages of meiosis would homologous chromosomes be paired?

- A) prophase of meiosis I
- B) prophase of meiosis II
- C) metaphase of meiosis I
- D) metaphase of meiosis II
- E) a and c are correct

Ans: E

Difficulty: 2

25. Which of the following events occurs during mitosis but not meiosis?

- A) Segregation of sister chromatids.
- B) Pairing of homologous chromosomes.
- C) Crossing-over between homologous chromosomes.
- D) Alignment of chromosomes on the metaphase plate.
- E) None of the above

Ans: E

Difficulty: 2

26. The mitotic stage during which chromosomes begin attaching to spindle fibers and moving randomly and reversibly to the centrosomes is:

- A) prophase.
- B) prometaphase.
- C) metaphase.
- D) anaphase.
- E) telophase.

Ans: B

Difficulty: 1

27. If a geneticist were to closely examine the make-up of a single autosomal chromosome from one of your cells, that chromosome would be found to be:

- A) derived entirely from genes from just one of your grandparents.
- B) a mosaic of genes derived from your mother and father.
- C) a mosaic of genes derived from all four of your grandparents.
- D) a mosaic of genes derived from just two of your grandparents — either your two grandfathers or your two grandmothers.
- E) a mosaic of genes derived from just two of your grandparents — either your maternal grandparents, or your paternal grandparents.

Ans: E

Difficulty: 4

28. If you are a male, your X chromosome contains genes derived from:

- A) your paternal grandfather only.
- B) your maternal grandfather only.
- C) both your paternal and maternal grandfathers.
- D) both your paternal grandfather and grandmother.
- E) both your maternal grandfather and grandmother.

Ans: E

Difficulty: 4

29. The chromosomal structure to which spindle fibers attach during the mitotic divisions is referred to as a:

- A) chromatid.
- B) centrosome.
- C) kinetochore.
- D) metaphase plate.
- E) centromere.

Ans: C

Difficulty: 1

30. Microtubules that attach to centrosomes but not chromosomes and that interdigitate with microtubules emanating from the centrosome at the opposite pole are known as:

- A) kinetochore microtubules.
- B) polar microtubules.
- C) astral microtubules.
- D) interdigitating microtubules.
- E) none of the above

Ans: B

Difficulty: 1

31. Mitosis results in _____ in chromosome number while meiosis results in _____ chromosome number.

- A) a doubling of, no change in
- B) no change in, no change in
- C) a reduction by half in, no change in
- D) no change in, a doubling of
- E) no change in, a reduction by half in

Ans: E

Difficulty: 2

32. Cells said to be in the G₀ stage:
- A) have two chromatids per chromosome.
 - B) are replicating their chromosomes.
 - C) are about to enter the mitotic phase of the cell cycle.
 - D) are in an extended G₁ phase and no longer cycling.
 - E) are dead.
- Ans: D
Difficulty: 1
33. Cells in G₂ stage of the cell cycle have _____ as cells of the same species in the G₁ stage.
- A) twice as many chromosomes
 - B) twice as many chromatids
 - C) half as many chromatids
 - D) the same number of chromatids
 - E) half as many chromosomes
- Ans: A
Difficulty: 1
34. A cross between a white-eyed female fruit fly and a red-eyed male would generate which of the following types of progeny?
- A) Red-eyed females and white-eyed males.
 - B) White-eyed females and red-eyed males.
 - C) All red-eyed females and a 50:50 mixture of white-eyed and red-eyed males.
 - D) All red-eyed males and a 50:50 mixture of white-eyed and red-eyed females.
 - E) The result can not be predicted because it depends on whether the female is homozygous or heterozygous.
- Ans: A
Difficulty: 2
35. A female fruit fly with singed bristles was mated with a male from a true-breeding wild type stock with long bristles. All of the F₁ females had wild-type bristles and all of the F₁ males had singed bristles. If the F₁ flies are intercrossed, the expected ratio of long to singed bristles in the F₂ flies is:
- A) 1:0 in both sexes (i.e. males and females will all have long bristles).
 - B) 3:1 in both sexes.
 - C) 3:1 in females, while all the males will have singed bristles.
 - D) 1:1 in females, while all the males will have singed bristles.
 - E) 1:1 in both sexes.
- Ans: E
Difficulty: 2

36. Red-green color-blindness is controlled by an X chromosomal gene in humans. A normal man and woman whose fathers are both color-blind marry. What is the probability that their first child will be color-blind?

- A) $1/2$
- B) $1/4$
- C) $1/8$
- D) $1/3$
- E) 0

Ans: B

Difficulty: 2

37. Hemophilia is caused by an X-linked mutation in humans. If a woman whose paternal uncle (father's brother) was a hemophiliac marries a man whose brother is also a hemophiliac, what is the probability of their first child having hemophilia (note: assume no other hemophilia in the pedigree and no hidden carriers).

- A) 0
- B) $1/4$
- C) $1/2$
- D) $1/8$
- E) 1

Ans: A

Difficulty: 3

38. A white-eyed female fruit fly is mated to a red-eyed male. One of their progeny is a white-eyed female. What is the likely karyotype of this white-eyed daughter?

- A) XX
- B) XXY
- C) XO
- D) XXX
- E) XYY

Ans: B

Difficulty: 2

39. A white-eyed female fruit fly is mated to a red-eyed male. One of their progeny is a white-eyed female. This white-eyed daughter likely arose from nondisjunction of the sex chromosomes during _____ in the _____.

- A) meiosis I, mother
- B) meiosis II, mother
- C) meiosis I, father
- D) meiosis II, father
- E) meiosis I or II, mother

Ans: E

Difficulty: 3

40. Males with one copy of an X-linked gene are said to be _____ for that gene.
A) homozygous
B) heteroallelic
C) heterozygous
D) hemizygous
E) deficient
Ans: D
Difficulty: 1
41. In chickens, it is the females that have two different sex chromosomes (Z and W) while the males have two Z chromosomes. A Z-linked gene controls the pattern of the feathers with the dominant B allele causing the barred pattern and the b allele causing non-barred feathers. From which of the following crosses would all of the daughters be of one type (barred or non-barred) and all of the sons the other type?
A) barred females × non-barred males
B) non-barred females × barred males
C) non-barred females × non-barred males
D) barred females × barred males
E) more than one of the above
Ans: A
Difficulty: 4
42. In *Drosophila*, the recessive allele for forked bristles is sex-linked, and the recessive allele for purple eyes is autosomal. The wild-type alleles are for straight bristles and red eyes, respectively. A female heterozygous for both pairs of alleles is mated with a male with forked bristles and purple eyes. What percentage of the daughters will have forked bristles and purple eyes?
A) 100%
B) 75%
C) 50%
D) 25%
E) 0%
Ans: D
Difficulty: 3
43. In what proportion of a man's sperm will all 23 chromosomes have centromeres inherited from the same parent (either his mother or his father)?
A) 1/2
B) $(1/2)^{23}$
C) $(1/2)^{46}$
D) $(1/2)^{22}$
E) $(1/4)^{46}$
Ans: D
Difficulty: 4

44. Suppose you discover a new species of worm that exists in two forms — slimy and non-slimy. You find that when slimy females are mated to non-slimy males the offspring consist of slimy males and non-slimy females, whereas when non-slimy females are mated with slimy males, the offspring of both sexes are all slimy. You would conclude that the _____ allele is dominant and that _____ are the heterogametic sex (the sex with two different sex chromosomes) in this species of worm.

- A) slimy, females
- B) slimy, males
- C) non-slimy, females
- D) non-slimy, males

Ans: A

Difficulty: 4

45. Which of the following is not a property of homologous chromosomes?

- A) They pair physically during prophase of meiosis I.
- B) They exchange genes by crossing over during meiosis.
- C) They carry alleles for the same genes at the same chromosomal position.
- D) They are attached to the same centromere during G2 of the cell cycle.
- E) They segregate to opposite poles at anaphase of meiosis I.

Ans: D

Difficulty: 2

46. Suppose you discover a new variant in which mice have spiky fur instead of the usual soft fur. You notice that this trait seems only to be present in males. To investigate this pattern, you cross a spiky-fur male with a soft-fur female, and find that all of the F1 progeny of both sexes have soft fur. You then interbreed the F1 and observe that all of the F2 females have soft fur, but 1/4 of the F2 males have spiky fur. You conclude that:

- A) the spiky allele is an X-linked recessive.
- B) the spiky allele is Y-linked.
- C) the spiky allele is autosomal recessive with sex-limited expression.
- D) the spiky allele is an X-linked dominant.
- E) the spiky allele is an autosomal dominant with sex-limited expression.

Ans: C

Difficulty: 3

47. In fruit flies, brown eyes can be caused by the recessive mutations *pn* (prune), *bw* (brown) or *ry* (rosy), which are on the X chromosome, second chromosome and third chromosome, respectively. Wild-type alleles for all three genes must be present for eyes to be red. Suppose that two brown-eyed flies are crossed and their progeny consist of brown-eyed sons and red-eyed daughters. Which mutation is responsible for the brown eyes in the parental female?

A) *pn*
B) *bw*
C) *ry*
D) either *bw* or *ry*
E) any of the three — *pn*, *bw* or *ry*

Ans: A

Difficulty: 4

48. Color vision depends upon the dominant alleles of three genes — the *R* gene and the *G* gene are both on the X chromosome, while the *B* gene is autosomal. Recessive mutations in any of the three genes can cause color-blindness. Suppose a colorblind man marries a colorblind woman and all of their offspring have normal vision. What is the genotype of the woman?

A) *RRGGbb*
B) *RRggBB*
C) *rrGGbb*
D) *RRggBB* or *RRGGbb*
E) *rrGGbb* or *rrggBB*

Ans: A

Difficulty: 4

49. A variable feature of some chromosomes in corn is the presence or absence of knobs at particular sites. Suppose that one member of each of two pairs of chromosomes in a corn plant has a knob. If this plant were crossed with a knobless plant, what percentage of the offspring is expected to be knobless?

A) 100%
B) 75%
C) 50%
D) 25%
E) 0%

Ans: D

Difficulty: 3

50. What aspect of chromosome behavior most clearly accounts for Mendel's law of segregation?
- A) Movement of sister chromatids to opposite poles at anaphase II of meiosis.
 - B) Movement of homologous chromosomes to opposite poles at anaphase I of meiosis.
 - C) Crossing over between homologous chromosomes during prophase I of meiosis.
 - D) Replication of chromosomes prior to meiosis.
 - E) Independent alignment of different homologous pairs on the metaphase I spindle.

Ans: B

Difficulty: 2

51. What aspect of chromosome behavior most clearly accounts for Mendel's law of independent assortment?
- A) Movement of sister chromatids to opposite poles at anaphase II of meiosis.
 - B) Movement of homologous chromosomes to opposite poles at anaphase I of meiosis.
 - C) Crossing over between homologous chromosomes during prophase I of meiosis.
 - D) Replication of chromosomes prior to meiosis.
 - E) Independent alignment of different homologous pairs on the metaphase I spindle.

Ans: E

Difficulty: 2

52. Which aspect(s) of chromosome behavior is/are primarily responsible for the tremendous amount of genetic variability associated with sexual reproduction?
- A) Segregation of sister chromatids at anaphase II of meiosis.
 - B) Segregation of homologous chromosomes at anaphase I of meiosis.
 - C) Crossing over between homologous chromosomes during prophase I of meiosis.
 - D) Independent alignment of different homologous pairs on the metaphase I spindle.
 - E) Both c and d contribute to the generation of variability.

Ans: E

Difficulty: 2

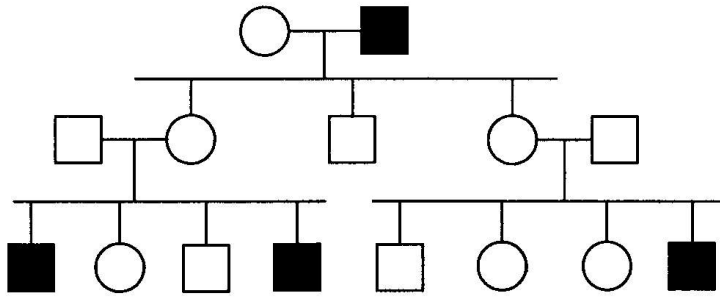
53. Which sex chromosomes are limited to only one sex?

- A) X and Z
- B) X and W
- C) Y and Z
- D) Y and W
- E) X and Y

Ans: D

Difficulty: 1

54. In the following pedigree, the indicated trait is caused by what type of allele?

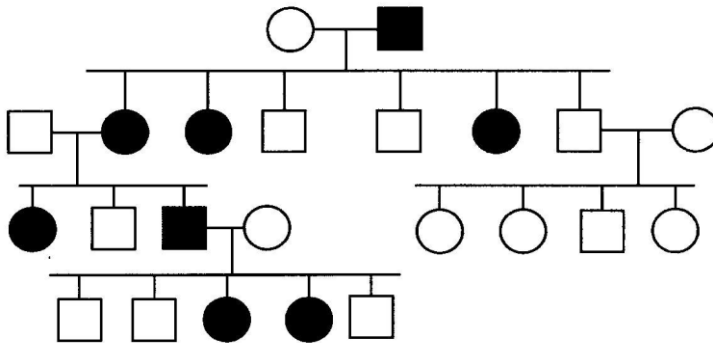


- A) autosomal recessive
- B) autosomal dominant
- C) X-linked recessive
- D) X-linked dominant
- E) Y-linked

Ans: C

Difficulty: 2

55. In the following pedigree, the indicated trait is caused by what type of allele?

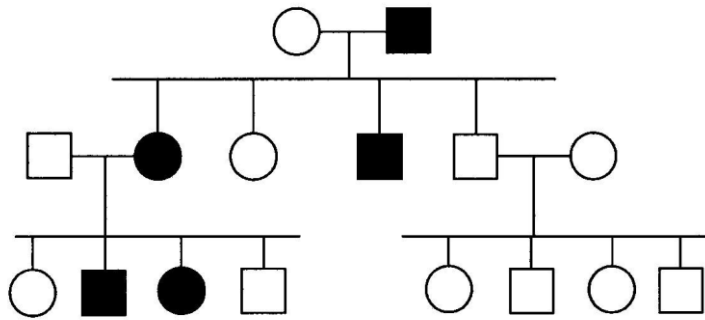


- A) autosomal recessive
- B) autosomal dominant
- C) X-linked recessive
- D) X-linked dominant
- E) Y-linked

Ans: D

Difficulty: 2

56. In the following pedigree, the indicated trait is caused by what type of allele?

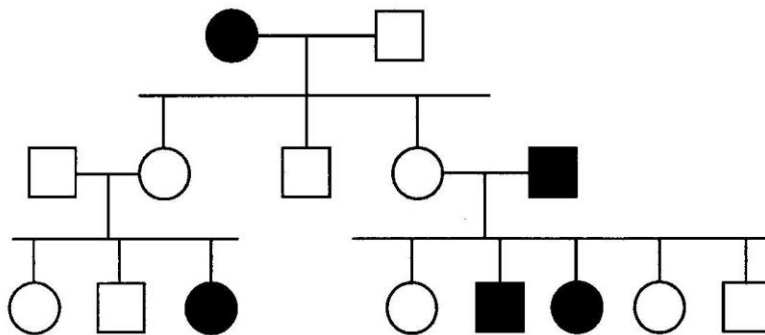


- A) autosomal recessive
- B) autosomal dominant
- C) X-linked recessive
- D) X-linked dominant
- E) Y-linked

Ans: B

Difficulty: 2

57. In the following pedigree, the indicated trait is caused by what type of allele?

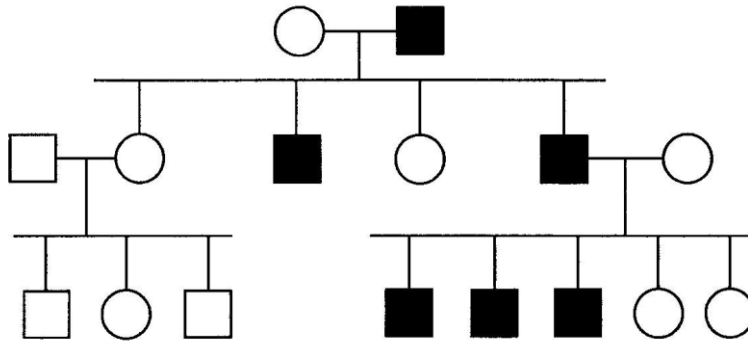


- A) autosomal recessive
- B) autosomal dominant
- C) X-linked recessive
- D) X-linked dominant
- E) Y-linked

Ans: A

Difficulty: 2

58. In the following pedigree, the indicated trait is caused by what type of allele?



- A) autosomal recessive
- B) autosomal dominant
- C) X-linked recessive
- D) X-linked dominant
- E) Y-linked

Ans: E

Difficulty: 2

59. A female fruit fly with vermilion eyes and normal wings is crossed to a male with normal red eyes and cut wings. The F1 progeny consist of females with red eyes and normal wings, and males with vermilion eyes and normal wings. When the F1 progeny are interbred, the F2 consists of two types of females — vermilion eyes, normal wings (1/2) and red eyes, normal wings (1/2), and two types of males — vermilion eyes, normal wings (1/2) and red eyes, cut wings (1/2). Are the genes for eye-color and wing-type X-linked or autosomal?

- A) The eye-color gene is X-linked and the wing-type gene is autosomal.
- B) The eye-color gene is autosomal and the wing-type gene is X-linked.
- C) Both genes are X-linked.
- D) Both genes are autosomal.
- E) More than one of the above is consistent with the data.

Ans: C

Difficulty: 4

60. In fruit flies, an X-linked dominant mutation (N) causes Notch wings in heterozygous females but is lethal in hemizygous or homozygous condition. What ratio of offspring would be observed in a cross of a Notch female by a normal male?

- A) 1/3 Notch females, 1/3 normal females, 1/3 normal males
- B) 1/4 Notch females, 1/4 normal females, 1/4 Notch males, 1/4 normal males
- C) 1/2 Notch females, 1/2 normal males
- D) 1/2 normal females, 1/2 Notch males
- E) 2/3 Notch females, 1/3 normal males

Ans: A

Difficulty: 3

61. The following is a list of mitotic events out of order.
- A) Chromosomes align on the midplate of the cell.
 - B) Kinetochores begin attaching to spindle fibers.
 - C) Nuclear membrane reforms, chromosomes decondense
 - D) Chromosomes condense, centrosomes migrate to opposite sides of nucleus.
 - E) Sister chromatids separate, move to opposite poles.

What is the correct order of events?

- A) BDACE
- B) DABEC
- C) DBAEC
- D) ABDCE
- E) EDBAC

Ans: C

Difficulty: 2

62. Here is a list of meiotic events out of order.
- A) Segregation of homologous chromosomes to opposite poles.
 - B) Segregation of sister chromatids to opposite poles.
 - C) Alignment of homologous pairs on the midplate of the cell.
 - D) Pairing and synapsis of homologous chromosomes.
 - E) Condensation of chromosomes in a diploid nucleus.

What is the correct order of events?

- A) CDEAB
- B) DCEBA
- C) EDCBA
- D) EDCAB
- E) DCABE

Ans: D

Difficulty: 2

63. A cell with three pairs of chromosomes has the genotype AaBbCc, such that each pair of alleles is on a different pair of chromosomes. If this cell were to undergo a meiotic division, how many genetically different types of gametes could be produced?

- A) 1
- B) 3
- C) 6
- D) 8
- E) 9

Ans: D

Difficulty: 2

64. A cell with three pairs of chromosomes has the genotype AaBbCc, such that each pair of alleles is on a different pair of chromosomes. If this cell were to undergo a mitotic division, how many genetically different types of gametes could be produced?

- A) 1
- B) 3
- C) 6
- D) 8
- E) 9

Ans: A

Difficulty: 2

65. In humans, XO individuals are females with Turner's syndrome and XXY individuals are males with Klinefelter's syndrome. Which of the following events could not give rise to a Klinefelter's male?

- A) Nondisjunction at meiosis I in the mother.
- B) Nondisjunction at meiosis II in the mother.
- C) Nondisjunction at meiosis I in the father.
- D) Nondisjunction at meiosis II in the father.
- E) All of the above could give rise to a Klinefelter's male.

Ans: D

Difficulty: 3

66. In humans, XO individuals are females with Turner's syndrome and XXY individuals are males with Klinefelter's syndrome. Red-green color blindness is caused by an X-linked recessive allele. Suppose a colorblind man and a normal woman with no family history of color-blindness married and had a daughter who was colorblind and had Turner's syndrome. Which event could have given rise to this offspring?

- A) Nondisjunction at meiosis I in the mother.
- B) Nondisjunction at meiosis II in the mother.
- C) Nondisjunction at meiosis I in the father.
- D) Nondisjunction at meiosis II in the father.
- E) Either a or b could give rise to such an offspring.

Ans: E

Difficulty: 3

67. Individuals with an XXY karyotype are _____ in humans and _____ in fruit flies.

- A) male, male
- B) male, female
- C) female, male
- D) female, female
- E) male, intersexual

Ans: B

Difficulty: 2

68. In animal gametogenesis, a single primary spermatocyte generates _____ sperm, while a single primary oocyte generates _____ egg(s).

- A) 1, 4
- B) 1, 1
- C) 4, 2
- D) 4, 1
- E) 4, 4

Ans: D

Difficulty: 1

Matching

69. Match the event with the appropriate stage of mitosis.

- | | |
|---|-----------------|
| 1. Sister chromatids separate and move to opposite poles. | a. prophase |
| 2. Chromosomes begin attaching to spindle fibers. | b. prometaphase |
| 3. Chromosomes decondense and nuclear membrane reforms. | c. metaphase |
| 4. Chromosomes condense and spindle forms. | d. anaphase |
| 5. Sister chromatids attached to opposite spindle poles. | e. telophase |

Ans: 1-b, 2-d, 3-e, 4-a, 5-c

Difficulty: 2

70. Match the chromosomal event with the appropriate stage of meiosis.

- | | |
|---|-----------------|
| 1. Homologous chromosomes pair and cross over. | a. prophase |
| 3. Homologous chromosomes segregate to opposite poles. | b. metaphase |
| 3. Sister chromatids segregate to opposite poles. | c. anaphase I |
| 4. Pairs of chromosomes align on the spindle. | d. metaphase II |
| 5. Sister chromatids attach to opposite poles of spindle. | e. anaphase II |

Ans: 1-a, 2-c, 3-e, 4-b, 5-d

Difficulty: 2

71. Match the event with the cell division process in which it occurs. (Note: there can be more than one correct letter for each blank.)

- | | |
|--|---------------|
| 1. Pairing of homologous chromosomes. | a. meiosis I |
| 2. Segregation of sister chromatids. | b. meiosis II |
| 3. Condensation and decondensation of chromosomes. | c. mitosis |
| 4. Segregation of homologous chromosomes. | |
| 5. Crossing over. | |

Ans: 1-a, 2-b,c, 3-a,b,c, 4-a, 5-a

Difficulty: 3

72. Match the event with the type of cell in which it occurs during spermatogenesis. (Note: a cell type may be used more than once or not at all, but there is only one correct letter per blank.)

- | | |
|--|----------------------------|
| 1. Homologous chromosomes pair and cross over. | a. Spermatogonia |
| 2. Sister chromatids segregate to opposite poles. | b. Primary spermatocytes |
| 3. Cells undergo morphogenetic changes. | c. Secondary spermatocytes |
| 4. Cells divide mitotically. | d. Spermatids |
| 5. Homologous chromosomes segregate to opposite poles. | |

Ans: 1-b, 2-c, 3-d, 4-a, 5-b

Difficulty: 2

73. Match the events with the appropriate stage of the cell cycle.

- | | |
|---|------------|
| 1. Chromosomes replicate. | a. G1 |
| 2. Growth period between replication and division, each chromosome has two sister chromatids. | b. S phase |
| 3. Growth period between division and replication, each chromosome has only one chromatid. | c. G2 |
| 4. An extended interphase in non-cycling cells. | d. M phase |
| 5. Cells divide; chromosomes undergo mitosis. | e. G0 |

Ans: 1-b, 2-c, 3-a, 4-e, 5-d

Difficulty: 2

74. Match the genotype with the appropriate term and with the appropriate sex (put a, b or c, and M or F on each blank).

<u>Genotype</u>	<u>Terms</u>	<u>Sex</u>
1. X^wX^w _____	a. homozygous	M
2. $X^{w+}X^w$ _____	b. heterozygous	F
3. $X^{w+}Y$ _____	c. hemizygous	
4. X^wO _____		
5. Z^BZ^B _____		
6. Z^bW _____		
7. Z^BZ^b _____		

Ans: 1-aF, 2-bF, 3-cM, 4-cM, 5-aM, 6-cF, 7-bM

Difficulty: 3

75. Match the stage of oogenesis in humans with the appropriate numbers of chromosomes and chromatids.

<u>Stage of Oogenesis</u>	<u># of chromosomes</u>	<u># of chromatids</u>
1. Primary oocytes _____	a. 23	d. 23
2. Secondary oocytes _____	b. 46	e. 46
3. Ootids _____	c. 92	f. 92
4. Ova _____		

Ans: 1-bf, 2-ae, 3-ad, 4-ad

Difficulty: 2

True or False

76. Sister chromatids are identical chromosomes, duplicated during metaphase and still attached to each other.
Ans: True
Difficulty: 1
77. The nonsex chromosomes are called autosomes.
Ans: True
Difficulty: 1
78. A female that is XXO has Turner Syndrome.
Ans: False
Difficulty: 2
79. A chromatin-laden nucleus is quiescent.
Ans: False
Difficulty: 2
80. The period between cell cycles is called interphase.
Ans: True
Difficulty: 1
81. During the S and G₂ stages of interphase, the centrosomes replicate, producing two centrosomes that remain in extremely close proximity.
Ans: True
Difficulty: 1
82. Chromosomes attach to these microtubules through linkage groups.
Ans: False
Difficulty: 1
83. Polar microtubules are short and unstable and extend out from the centrosome toward the cell's periphery.
Ans: False
Difficulty: 2
84. A somatic cell is a cell destined for a specialized role in the production of gametes.
Ans: False
Difficulty: 1

85. During pachytene, structures called recombination nodules begin to appear along the synaptonemal complex, and an exchange of parts between nonsister chromatids occurs at these nodules.

Ans: True

Difficulty: 2

Short Answer

86. In humans, why is pattern baldness seen only in men?

Ans: Pattern baldness is an example of sex-limited expression. It is caused by heterozygosity for an autosomal baldness gene, which when homozygous causes baldness in both sexes; in the heterozygous condition, pattern baldness is expressed only in males and is thus sex-limited.

Difficulty: 2

87. Give examples of three different methods by which chromosomes can determine sex.

Ans: XX females, XY males — humans, fruit flies; XX females, XO males — some moths and other insects; ZZ males, ZW females — birds and butterflies.

Difficulty: 2

88. What are the consequences of a mutation that disrupts cytokinesis for cells that are actively dividing?

Ans: Mitosis without cytokinesis leads to multinucleate cells with multiple copies of each chromosome.

Difficulty: 1

89. What is the major cause of Down's syndrome?

Ans: Trisomy (three copies) of chromosome 21, which results from nondisjunction of chromosome 21 in meiosis of the mother (80% of cases) or father (20% of cases).

Difficulty: 1

90. What is meant by the term “checkpoint” with respect to regulation of the cell cycle?

Ans: A checkpoint is a moment when cell cycle progress can be stopped temporarily if a preceding step is not yet complete.

Difficulty: 2

91. What was the evidence that led Walter Sutton to conclude that the X and Y chromosomes determine sex?

Ans: Sutton studied the chromosomes of grasshoppers and showed that males always had one X and one Y while females always had two Xs.

Difficulty: 2

92. Bridges crossed white-eyed female and red-eyed male fruit flies and recovered rare exceptional white-eyed females and red-eyed males in addition to the usual red-eyed females and white-eyed males. How did Bridges' observations lend support to the chromosome theory of inheritance?

Ans: Bridges showed that the exceptional flies also had exceptional karyotypes for the sex chromosomes (XO males and XXY females). So, a reversal in the normal pattern of transmission of a genetic trait is perfectly correlated with a mistake in the pattern of transmission of a particular pair of chromosomes (the sex chromosomes), such that the exceptional sons inherit their X (and their eye color) from their father, and the exceptional daughters inherit both of their Xs (and their eye color) from their mother, proving that the eye color gene must be on the X chromosome.

Difficulty: 4

93. What are two important functions of meiosis?

Ans: To reduce chromosome number by half in preparation for fertilization so that biparental inheritance does not lead to an increase in chromosome number; to generate offspring that differ genetically from their parents and from each other.

Difficulty: 3

94. Why are recessive X-linked traits more commonly seen in men than in women?

Ans: Because expression of an X-linked trait in a man requires only that he inherit one copy of the allele from his mother, whereas expression in females requires inheritance of two copies, one from each parent.

Difficulty: 2

95. Why are dominant X-linked traits seen in women more commonly than in men?

Ans: There are two reasons: 1) Any X chromosome is twice as likely to be present in a woman as a man because women have two X chromosomes and men have one; 2) many dominant mutations are recessive lethals, so that for many dominant X traits, men that inherit them die during development because of hemizogosity.

Difficulty: 3

Experimental Design and Interpretation of Data

96. In *Drosophila*, vestigial wings (vg) is a recessive autosomal mutant allele, and yellow body (y) is a recessive X-linked mutant allele. The wild-type alleles, vg^+ and y^+ control normal wings and dark-brown body color respectively. If a homozygous yellow, vestigial female is crossed to a normal male and the F1 offspring are interbred, what will be the phenotypes and ratios of the F1 and F2?

Ans: F1 — females: all brown body, normal wings
males: all yellow body, normal wings
F2 — females and males: brown body, normal wings — 3/8
brown body, vestigial wings — 1/8
yellow body, normal wings — 3/8
yellow body, vestigial wings — 1/8

Difficulty: 4

97. In *Drosophila*, white eyes (w) and yellow body (y) are both recessive X-linked mutations. The wild type alleles, w^+ and y^+ , control red eyes and dark body color, respectively. If a homozygous yellow body, red eyed female is crossed with a dark body, white-eyed male, and F1 progeny are interbred, what will the phenotypes and ratios of the F1 and F2 be?

Ans: F1 — females: all dark body, red eyes
males: all yellow body, red eyes
F2 — females: yellow body, red eyes — 1/2
dark body, red eyes — 1/2
males: yellow body, red eyes — 1/2
dark body, white eyes — 1/2

Difficulty: 4

98. Suppose you discovered a new mutation in mice that caused a curved spine. You notice that this mutation is only seen in females. When curved-spine females are crossed with normal males, the progeny are always recovered in the ratio: 1/3 curved-spine females, 1/3 normal females, 1/3 normal males. Explain the genetic basis for this ratio.

Ans: The gene is X-linked; the curved-spine allele is dominant to the normal allele in heterozygotes but causes lethality in hemizygous males.

Difficulty: 4

99. In crosses of white-eyed *Drosophila* females by red-eyed males, Bridges recovered white-eyed daughters and red-eyed sons at a rate of around one per 2,000 offspring. (Most of the offspring were white-eyed males and red-eyed females.) He hypothesized that these exceptional progeny resulted from nondisjunction of the X chromosomes in meiosis in the female. Why did he suspect that nondisjunction was occurring in the female parent? What types of progeny would result from nondisjunction in the male parent?

Ans: XXY red-eyed females and XO white-eyed males, so male nondisjunction does not account for the observed exceptions.

Difficulty: 3

100. In crosses of white-eyed *Drosophila* females by red-eyed males, Bridges recovered white-eyed daughters and red-eyed sons at a rate of around one per 2,000 offspring. (Most of the offspring were white-eyed males and red-eyed females.) He hypothesized that these exceptional progeny resulted from nondisjunction of the X chromosomes in meiosis in the female. What is the expected karyotype of the exceptional red-eyed males? Give two ways to test this predicted karyotype.